Task 01: Networking Fundamentals, Nmap Scanning, and Automation Scripting

Organization: CYART
Task Title: Networking Fundamentals, Nmap Scanning, and Automation Scripting
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Introduction

Clear and structured documentation is a critical part of cybersecurity and network engineering workflows. Proper documentation not only helps in understanding technical processes but also ensures that findings can be reviewed, audited, and reused in the future.

This document demonstrates practical understanding of **networking fundamentals**, **network scanning using Nmap**, and **automation using Python scripting**. Visual elements such as tables and a network diagram are included to improve clarity and comprehension.
The intended audience for this document is **internal technical teams and mentors**, serving as a technical reference for learning and evaluation.

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1. Overview of the Task

The objective of this task is to:

- Understand basic networking concepts such as IP addresses, ports, and protocols.
- Perform different types of network scans using **Nmap**.
- Analyze scan results to identify open ports, running services, and potential security risks.
- Automate the scanning process using **Python** and the `python-nmap` library.
- Present findings in a clear, well-structured, and professional manner.

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2. Networking Fundamentals

2.1 IP Address (Internet Protocol Address)

An **IP address** is a unique numerical identifier assigned to a device on a network. It allows devices to locate and communicate with each other.

Example:
`127.0.0.1` – This address refers to **localhost**, which represents the local machine itself.

2.2 Port

A **port** is a logical communication endpoint that allows different services to run on the same IP address.

Examples of common ports:

- **22** – Secure Shell (SSH)
- **80** – Hypertext Transfer Protocol (HTTP)
- **443** – HTTPS (Secure Web)

2.3 Protocols

- **TCP (Transmission Control Protocol):**
  A connection-oriented protocol that ensures reliable data transmission.
- **UDP (User Datagram Protocol):**
  A connectionless protocol that prioritizes speed over reliability.

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3. Target Description

- **Target System:** Local Machine
- **IP Address:** 127.0.0.1 (Localhost)
- **Operating System:** Windows
- **Reason for Selection:**
  The localhost environment is a safe and authorized target for testing and learning, ensuring no legal or ethical violations.

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4. Tools Used

- **Nmap (Network Mapper):**
  Used for network discovery and port scanning.
- **Npcap:**
  Packet capture driver required for advanced scans such as SYN scans.
- **Python:**
  Used for automating the scanning process.
- **python-nmap Library:**
  A Python wrapper that enables programmatic interaction with Nmap.

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5. Nmap Scans Performed

Figure 1: Verifying Nmap installation and version

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.26100.7628]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32> nmap -v
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-02 16:13 +0530
Read data files from: C:\Program Files (x86)\Nmap
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.04 seconds
           Raw packets sent: 0 (0B) | Rcvd: 0 (0B)

C:\Windows\System32> nmap -sS 127.0.0.1 -oN syn_scan.txt
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-02 16:14 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00075s latency).
Not shown: 994 closed tcp ports (reset)
PORT     STATE SERVICE
```

5.1 SYN Scan

```
nmap -sS 127.0.0.1 -oN syn_scan.txt
```

Figure 2: SYN scan execution and output

```
               2 Dir(s)   7,680,704,512 bytes free

C:\Users\Tanish\Desktop\Task01_Nmap_Networking>nmap -sT 127.0.0.1 -oN tcp_scan.txt
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-02 16:27 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00033s latency).
Not shown: 994 closed tcp ports (conn-refused)
PORT      STATE SERVICE
135/tcp   open  msrpc
445/tcp   open  microsoft-ds
902/tcp   open  iss-realsecure
912/tcp   open  apex-mesh
5357/tcp open  wsdapi
8000/tcp open  http-alt

Nmap done: 1 IP address (1 host up) scanned in 0.34 seconds

C:\Users\Tanish\Desktop\Task01_Nmap_Networking>dir
 Volume in drive C is Windows
 Volume Serial Number is 2A25-A372

 Directory of C:\Users\Tanish\Desktop\Task01_Nmap_Networking

02-02-2026  16:27    <DIR>          .
02-02-2026  16:18    <DIR>          ..
02-02-2026  16:25               485 syn_scan.txt
02-02-2026  16:27               492 tcp_scan.txt
```

- Uses a half-open TCP handshake.
- Faster and stealthier than full TCP scans.

5.2 TCP Connect Scan

```
nmap -sT 127.0.0.1 -oN tcp_scan.txt
```

- Completes the full TCP three-way handshake.
- More detectable but reliable.

5.3 UDP Scan

Figure 3: Generated Nmap output files

```
nmap -sU 127.0.0.1 -oN udp_scan.txt
```

- Scans UDP-based services.
- Slower due to the connectionless nature of UDP.

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6. Scan Results Summary

Port	Protocol	Service Name
135	TCP	MSRPC
445	TCP	Microsoft-DS (SMB)
902	TCP	ISS RealSecure
912	TCP	Apex Mesh
5357	TCP	WSDAPI
8000	TCP	HTTP-ALT

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7. Security Analysis

- **Port 445 (SMB):**
  If misconfigured, it can allow unauthorized file access and is commonly targeted by ransomware attacks.
- **Port 135 (MSRPC):**
  Vulnerable to remote procedure call exploits in unpatched systems.
- **Port 8000 (HTTP-ALT):**
  May expose web applications or administrative interfaces if not properly secured.

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8. SYN Scan vs TCP Connect Scan Comparison

Feature	SYN Scan	TCP Connect Scan
Handshake	Partial	Full
Stealth	High	Low
Speed	Faster	Slower
Detectability	Low	High

Both scans identified the same open ports; however, the SYN scan is preferred in security assessments due to its stealthy nature.

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9. UDP Scan Observations

The UDP scan detected limited or no open ports. This is expected behavior because:

- UDP does not use handshakes.
- Firewalls often filter or block UDP probes.
- Lack of response does not always indicate a closed port.

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10. Python Automation

A Python script (`nmap_automation.py`) was developed using the `python-nmap` library to automate the SYN scan process.

Key Features of the Script:

- Accepts target IP or hostname as user input.
- Performs a SYN scan programmatically.
- Extracts open ports and service names.
- Generates a structured output file (`scan_report.txt`) with a timestamp.

This automation demonstrates the ability to integrate security tools with scripting for efficiency.

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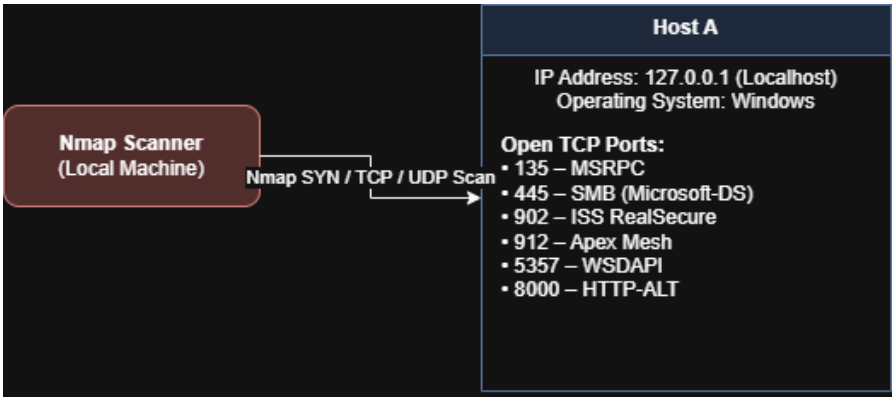
11. Network Diagram

Figure 5: Network diagram showing the localhost (127.0.0.1), detected open ports, and services.



The following diagram represents the scanned localhost system along with detected open ports and services.

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12. Key Learnings

- Practical understanding of TCP and UDP scanning techniques.
- Hands-on experience with Nmap and interpreting scan results.
- Awareness of security risks associated with open ports.
- Automation of security tasks using Python scripting.
- Importance of clear and structured technical documentation.

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13. Conclusion

This task provided practical exposure to network scanning and automation fundamentals. By combining Nmap with Python, the scanning process was automated efficiently while

maintaining clarity in documentation and reporting. The task successfully strengthened foundational skills required for cybersecurity and network engineering roles.

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14. References

- Nmap Official Documentation – https://nmap.org
- CYART Documentation Format Guidelines